Command Center

SAMPLE DATA

Flood scenario: what is happening, what is coming, and what to do.

Flood | Severity: High | 72 hours | Food 40,000 | Water 40,000 | Medical 2,500

EDIT SCENARIO | RUN AI ANALYSIS

1 Situation → 2 Prediction & priority → 3 Allocation → 4 Actions → What-if test

AI SUMMARY

3 of 7 areas need priority attention: E (critical), F and D (high).

Water and food cover about 56% of predicted need. The plan cuts unmet need in these 3 areas from 37,499 to 18,500 units.

APPLY RECOMMENDED PLAN

1 What is happening?

Current situation from your data.

TOTAL AFFECTED	CRITICAL / HIGH AREAS	TOTAL SHORTAGE
22,455	**1 / 2**	**64,444**
People in 7 areas	Priority score above threshold	Units still unmet after plan

PREDICTED HIGH-RISK	RESOURCES AVAILABLE	AVG COVERAGE
3	**82,500**	**61%**
Areas the model flags	Food + water + medical units	Average of 3 resources

RESOURCE COVERAGE (supply / predicted need)

Food	56%
Water	56%
Medical	73%

WARNING — Water coverage below 60%

2 Which areas need attention first?

AI predicts how many people will be affected, then ranks areas by priority score.

AREA RANKING

PREDICTION

RANK	AREA	AFFECTED NOW → PREDICTED	CHANGE	RISK	PRIORITY SCORE
#1	Area E	4,176 → **6,500**	+56%	CRITICAL	0.91
#2	Area F	3,800 → **5,000**	+32%	HIGH	0.74
#3	Area D	4,000 → **5,200**	+30%	HIGH	0.68
#4	Area C	3,300 → **3,900**	+18%	MEDIUM	0.51
#5	Area B	2,900 → **3,400**	+17%	MEDIUM	0.42
#6	Area G	2,429 → **2,600**	+7%	LOW	0.22
#7	Area A	1,850 → **2,100**	+14%	LOW	0.18

3 What should be sent where?

Optimizer splits supply across areas. Each cell: units sent, of units needed (coverage).

RECOMMENDED ALLOCATION

OPTIMIZATION

AREA	RISK	FOOD	WATER	MEDICAL
Area E	CRITICAL	16,250 of 16,250 (100%)	16,250 of 16,250 (100%)	780 of 780 (100%)
Area F	HIGH	12,500 of 12,500 (100%)	12,500 of 12,500 (100%)	600 of 600 (100%)

IMPACT OF OPTIMIZATION

UNMET NEED IN CRITICAL + HIGH AREAS

Equal-share baseline	37,499
AI-optimized plan	18,500

−51%

Reduction vs baseline. Every area keeps a 25% minimum

Area D	HIGH	3,750 of 13,000 (29%)	3,750 of 13,000 (29%)	624 of 624 (100%)
Area C	MEDIUM	2,438 of 9,750 (25%)	2,438 of 9,750 (25%)	253 of 468 (54%)
Area B	MEDIUM	2,125 of 8,500 (25%)	2,125 of 8,500 (25%)	102 of 408 (25%)
Area G	LOW	1,625 of 6,500 (25%)	1,625 of 6,500 (25%)	78 of 312 (25%)
Area A	LOW	1,312 of 5,250 (25%)	1,312 of 5,250 (25%)	63 of 252 (25%)

coverage.

4 What should we do now?

Top three recommended actions. Final decision rests with the emergency team.

CRITICAL **Area E**

Send 16,250 water, 16,250 food, 780 medical kits

Why: Affected people expected to rise 56%; water coverage 56% under equal share.
Effect: water coverage 56% → 100%

HIGH **Area F**

Send 12,500 water, 12,500 food, 600 medical kits

Why: Affected people expected to rise 32%; water coverage 56% under equal share.
Effect: water coverage 56% → 100%

HIGH **Area D**

Send 3,750 water, 3,750 food, 624 medical kits

Why: Affected people expected to rise 30%; water coverage 56% under equal share.
Effect: water coverage 56% → 29%

Monitor only: Area A, Area G, Area B (lower priority; minimum coverage guaranteed).

Decision-support tool; final decisions rest with emergency managers. Sample numbers are illustrative and are replaced by your data and model output.

Predictions

SAMPLE DATA

Which areas are likely to become critical, and by how much?

PREDICTED AFFECTED BY AREA

PREDICTION

AREA	NOW	PREDICTED	CHANGE	RISK
Area E	4,176	6,500	+56%	CRITICAL
Area F	3,800	5,000	+32%	HIGH
Area D	4,000	5,200	+30%	HIGH
Area C	3,300	3,900	+18%	MEDIUM
Area B	2,900	3,400	+17%	MEDIUM
Area G	2,429	2,600	+7%	LOW
Area A	1,850	2,100	+14%	LOW

HOW TO READ THIS

Predicted values come from the trained model. Risk level comes from the priority score thresholds (adjustable). A probability column appears only when the chosen model provides one.

CRITICAL	score 0.85 or more
HIGH	0.60 to 0.84
MEDIUM	0.35 to 0.59
LOW	below 0.35

CURRENT VS PREDICTED AFFECTED PEOPLE



Blue = now. Second bar = predicted (red when critical).

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Resource Optimizer

SAMPLE DATA

How should limited supplies be split to protect the highest-priority areas?

UNMET NEED, BASELINE	UNMET NEED, AI PLAN	REDUCTION	SOLVER STATUS
37,499	**18,500**	**51%**	OPTIMAL
Critical + high areas, equal share	Same areas, optimized	(before - after) / before	Shown from the solver result

ALLOCATION BY AREA (units sent, of units needed)

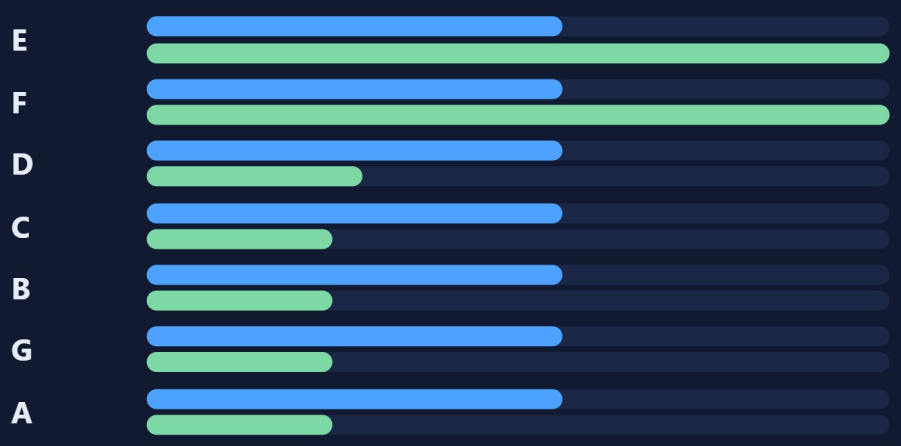
OPTIMIZATION

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OBJECTIVE AND RULES

Goal	**Minimise priority-weighted unmet need**
Rule 1	**Sent ≤ supply**
Rule 2	**Sent ≤ area need**
Rule 3	**Min. 25% coverage each area**
Sources	**No warehouse data**

WATER COVERAGE BY AREA: BASELINE VS AI PLAN



Blue = baseline. Green = AI plan.

Decision-support tool; final decisions rest with emergency managers. Sample numbers are illustrative and are replaced by your data and model output.

Decision Map

SAMPLE DATA

Click an area to see its risk, shortage and recommended action.

RISK MAP (layout is illustrative; real map uses coordinates from your data)

Area E
CRITICAL
6,500 predicted

Area F
HIGH
5,000 predicted

Area D
HIGH
5,200 predicted

Area C
MEDIUM
3,900 predicted

Area B
MEDIUM
3,400 predicted

Area G
LOW
2,600 predicted

Area A
LOW
2,100 predicted

CRITICAL HIGH MEDIUM LOW

AREA E

RECOMMENDATION

CRITICAL

Affected now	**4,176**
Predicted	**6,500**
Water needed	**16,250**
Water sent	**16,250**
Medical sent	**780**
Priority score	**0.91**

Top reasons: rising population, water shortage, high severity.

WHY? SIMULATE APPLY

Decision-support tool; final decisions rest with emergency managers. Sample numbers are illustrative and are replaced by your data and model output.

AI Explainability

SAMPLE DATA

Why did the model rate each area this way?

GLOBAL: WHAT MATTERS MOST (sample SHAP)

SHAP

Predicted affected population	0.31
Water shortage	0.24
Medical demand	0.18
Severity	0.15
Accessibility	0.07

WHY IS AREA E CRITICAL? (sample SHAP)

SHAP

Predicted affected population	+0.31
Water shortage	+0.24
Medical demand	+0.18
Severity	+0.15
Accessibility	-0.07

Positive values raise predicted risk; negative values lower it.

IN PLAIN WORDS

Area E is rated CRITICAL mainly because its affected population is predicted to grow and water supply is far below need.

These values describe how the model behaves. They do not prove cause and effect.

PRIORITY SCORE BREAKDOWN (weights are assumptions)

PART	WEIGHT	SCORE
Impact	0.30	0.28
Predicted demand	0.25	0.24
Shortage	0.25	0.23
Medical need	0.15	0.12
Access penalty	0.05	0.04

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What-If Simulator

SAMPLE DATA

Change conditions, rerun the pipeline, and compare plans.

CHANGE THE SCENARIO

SIMULATION

Water supply: **-30%**

Food supply: **0%**

Medical supply: **0%**

Affected population: **0%**

SIMULATE

BEFORE VS AFTER: WATER -30%

SIMULATION

MEASURE	CURRENT	AFTER	CHANGE
Water coverage	56%	39%	-17 pts
Water shortage	31,750	43,750	+12,000
Total shortage	64,444	76,444	+12,000
Unmet need, critical + high	18,500	30,500	+12,000
Water sent to Area E	16,250	4,750	-11,500

Calculated by rerunning the same optimizer on the changed supply.

SCENARIO A VS SCENARIO B

	A: CURRENT	B: WATER -30%	DIFFERENCE
Water available	40,000	28,000	-12,000
Water coverage	56%	39%	-17 pts
Total shortage	64,444	76,444	+12,000

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Model Performance

SAMPLE DATA

Is the model reliable? Evidence from held-out test data (sample numbers).

PRECISION	RECALL	F1	PR-AUC
0.84	**0.81**	**0.83**	**0.88**
Flagged areas that were truly high-risk	High-risk areas the model caught	Balance of precision and recall	Preferred when classes are imbalanced

MODEL COMPARISON (best chosen by F1)

MODEL	F1	PR-AUC	RESULT
Logistic Regression	0.74	0.79	
Random Forest	0.80	0.85	
XGBoost	0.83	0.88	BEST

CONFUSION MATRIX

	Pred. high	Pred. low
Actual high	38	9
Actual low	7	86

TRAINING SETUP (sample)

Split	**Stratified, 80/20**
Validation	**5-fold cross-validation**
Random seed	**42 (fixed)**
Note	**Accuracy alone is not used**

Decision-support tool; final decisions rest with emergency managers. Sample numbers are illustrative and are replaced by your data and model output.

AI Decision Audit

SAMPLE DATA

Every analysis is recorded so decisions can be reviewed.

DECISION LOG (sample entries)

AUDIT

TIME	SCENARIO	MODEL	TOP FEATURES	SOLVER	DETAILS
2026-10-08 10:12	Flood, High, base supply	XGBoost	Predicted pop., water shortage	optimal	OPEN
2026-10-08 10:20	Water -30%	XGBoost	Water shortage, severity	optimal	OPEN
2026-10-08 10:31	Population +20%	XGBoost	Predicted pop., medical demand	optimal	OPEN

DOWNLOAD CSV

EACH ENTRY RECORDS

Scenario inputs, model used, predictions, priority scores, allocation, main influencing features, objective, solver status and timestamp.

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